exit the room, so it breaks the problem down further to “move toward the door” or “move away from the door.” Since the door is closer to the dock, and there is a path from the door to the dock, the robot knows it needs to move toward the door, but again it doesn’t know what sequence of primitive actions will get it toward the door. Lastly, it needs to decide whether to move left, right, forward, or reverse. It can see the door is in front of it, so it moves forward. It keeps this process up until it exits the room, when it must do some more goal decomposition until it gets to the dock.

This is the essence of dynamic programming. It is a generic approach for solving certain kinds of problems that can be broken down into subproblems and sub-subproblems, and it has applications across many fields including bioinformatics, economics, and computer science.

In order to apply Bellman’s dynamic programming, we have to be able to break our problem into subproblems that we know how to solve. But even this seemingly innocuous assumption is difficult to realize in the real world. How do you break the high-level goal for a self-driving car of “get to point B from point A without crashing” into small non-crashing subproblems? Does a child learn to walk by first solving easier sub-walking problems? In RL, where we often have nuanced situations that may include some element of randomness, we can’t apply dynamic programming exactly as Bellman laid it out. In fact, DP can be considered one extreme of a continuum of problem-solving techniques, where the other end would be random trial and error.

Another way to view this learning continuum is that in some situations we have maximal knowledge of the environment and in others we have minimal knowledge of the environment, and we need to employ different strategies in each case. If you need to use the bathroom in your own house, you know exactly (well, unconsciously at least) what sequence of muscle movements will get you to the bathroom from any starting position (i.e., dynamic programming-ish). This is because you know your house extremely well—you have a more or less perfect *model* of your house in your mind. If you go to a party at a house that you’ve never been to before, you might have to look around until you find the bathroom on your own